Proof summaries

How to use. For each theorem, the bullet points are the "story" of the proof: (i) main tool/lemma, (ii) key decomposition/induction, (iii) conclusion/identification.

Theorem 1.6 (Conditional expectation existence)

Proof idea.

- Define a finite measure Q on $(\Omega, \mathcal{F})$ by $Q(F) = \int_F X \, d\mathbb{P}$ (first for $X \geq 0$).
- Note $Q \ll \mathbb{P}|_{\mathcal{F}}$, so Radon–Nikodym gives an $\mathcal{F}$-measurable density $Z = \frac{dQ}{d\mathbb{P}|_{\mathcal{F}}}$ with $\int_F Z \, d\mathbb{P} = \int_F X \, d\mathbb{P}$ for all $F \in \mathcal{F}$.
- For general $X \in L^1$, apply the nonnegative case to X^+ and X^- and set $Z = Z^+ - Z^-$.
- Uniqueness: if Z, Z' both work, test on $\{Z - Z' > 1/n\} \in \mathcal{F}$ to force $\mathbb{P}(Z > Z') = \mathbb{P}(Z' < Z) = 0$.

Theorem 1.8 (Properties of conditional expectation)

Proof idea.

- Fix versions $X_0 = \mathbb{E}[X \mid \mathcal{F}]$, $Y_0 = \mathbb{E}[Y \mid \mathcal{F}]$ and always prove identities by checking the defining property on every $F \in \mathcal{F}$ and then invoking a.s. uniqueness.
- Linearity and constants: build the candidate (e.g. $aX_0 + bY_0$ or the constant c), verify it is $\mathcal{F}$-measurable, and match integrals over $F \in \mathcal{F}$.
- Monotonicity: consider $A = \{X_0 > Y_0\} \in \mathcal{F}$ and integrate $Y_0 - X_0$ over A; the sign contradiction forces $\mathbb{P}(A) = 0$.
- Tower property: test on $G \in \mathcal{G} \subset \mathcal{F}$ and use the defining identity twice.
- Measurable factor / independence: prove first for indicators $Z = \mathbf{1}_{F_0}$, extend to simple functions and then to general Z by dominated convergence; for independence test $\int_F X \, d\mathbb{P} = \mathbb{E}[X]\mathbb{P}(F)$.

Theorem 1.9 (Best approximation / orthogonal projection)

Proof idea.

- Let $X_0 = \mathbb{E}[X \mid \mathcal{F}]$ and take any $Y \in L^2(\Omega, \mathcal{F}, \mathbb{P})$.
- Show orthogonality: $\mathbb{E}[(X - X_0)(X_0 - Y)] = \mathbb{E}\big[\mathbb{E}[X - X_0 \mid \mathcal{F}](X_0 - Y)\big] = 0$ (tower + $\mathcal{F}$-measurable factor).
- Apply Pythagoras in L^2: $\|X - Y\|_2^2 = \|X - X_0\|_2^2 + \|X_0 - Y\|_2^2 \geq \|X - X_0\|_2^2$.
- Equality iff $\|X_0 - Y\|_2 = 0$, i.e. $Y = X_0$ a.s., so X_0 is the orthogonal projection / best L^2 predictor.

Theorem 1.10 (Jensen inequality)

Proof idea.

- Use the supporting-line characterization of convexity: for each x_0, $g(x) \geq g(x_0) + m_{x_0}(x - x_0)$ for some slope m_{x_0} (e.g. right derivative).

- Choose a countable set of supporting lines (take $x_0 \in I \cap \mathbb{Q}$) so that g is the pointwise supremum of these affine functions.
- Insert X and take $\mathbb{E}[\cdot \mid \mathcal{F}]$: conditional expectation preserves affine functions, so each supporting-line inequality becomes an inequality involving $\mathbb{E}[X \mid \mathcal{F}]$.
- Take the supremum over the countable family (so no measurability issues beyond a single null set) to get $g(\mathbb{E}[X \mid \mathcal{F}]) \leq \mathbb{E}[g(X) \mid \mathcal{F}]$ a.s.

Theorem 2.4 (Existence of an RCD for real-valued random variables)

Proof idea.

- For each rational r, pick a version $F(r, \omega) = \mathbb{P}(Y \leq r \mid \mathcal{F})(\omega)$; these are $\mathcal{F}$-measurable numbers in $[0, 1]$.
- Use monotonicity of conditional expectation to enforce $r \mapsto F(r, \omega)$ nondecreasing outside one null set (take a countable union over $r \leq s$ in $\mathbb{Q}$).
- Force right limits and boundary conditions ($F(-n) \downarrow 0$, $F(n) \uparrow 1$) outside another null set using dominated convergence.
- Define for all real z the right-continuous hull $\tilde{F}(z, \omega) = \inf\{F(r, \omega) : r \in \mathbb{Q}, r > z\}$; for each good ω this is a genuine cdf.
- Let $\vartheta(\omega, \cdot)$ be the probability measure with cdf $\tilde{F}(\cdot, \omega)$; measurability on the generating class $(-\infty, r]$ and a π–λ (monotone class) argument give measurability for all Borel sets.
- Verify the defining identity first on $(-\infty, r]$ and extend to all $B \in \mathcal{B}(\mathbb{R})$ by uniqueness of measures on generators.

Theorem 2.8 (Polish $\implies$ Borel space)

Proof idea.

- Use that a Polish space has a countable base (separability) and nice limit behavior (completeness), which yields a standard Borel structure.
- Concretely, one can embed E as a Borel subset of a universal Polish space (e.g. $[0, 1]^{\mathbb{N}}$) via a Borel isomorphism built from a countable base / separating family.
- Thus $(E, \mathcal{B}(E))$ is measurably isomorphic to a Borel subset of $\mathbb{R}$, i.e. a Borel (standard Borel) space.

Theorem 2.10 (Existence of RCDs for Borel-valued random variables)

Proof idea.

- Take a measurable isomorphism $\xi : (E, \mathcal{E}) \to (B, \mathcal{B}(B))$ with $B \subset \mathbb{R}$ Borel (exists since E is a Borel space).
- Push Y forward to a real-valued variable $\tilde{Y} = \xi \circ Y$ and apply the real-valued existence theorem to get an RCD $\vartheta_{\tilde{Y},\mathcal{F}}$.
- Pull the kernel back: define $\vartheta_{Y,\mathcal{F}}(\omega, A) = \vartheta_{\tilde{Y},\mathcal{F}}(\omega, \xi(A))$; measurability and the defining identity transfer through the isomorphism.

Theorem 2.11 (Integral representation of conditional expectation)

Proof idea.

- First prove for simple $f = \sum_i \alpha_i \mathbf{1}_{A_i}$ using the RCD property: $\mathbb{E}[f(X)\mathbf{1}_B] = \sum_i \alpha_i \mathbb{P}(B \cap \{X \in A_i\}) = \int_B \int f \, d\vartheta_{X,\mathcal{F}} \, d\mathbb{P}$.
- Approximate any nonnegative f by an increasing sequence of simple functions and pass to the limit via monotone convergence (both inside the kernel integral and in expectation).
- Recognize the resulting $\mathcal{F}$-measurable function $\omega \mapsto \int f \, d\vartheta_{X,\mathcal{F}}(\omega, \cdot)$ as the unique version of $\mathbb{E}[f(X) \mid \mathcal{F}]$ by testing against indicators of $B \in \mathcal{F}$.

Theorem 2.12 (Disintegration lemma for probability measures on products)

Proof idea.

- Work on the probability space $(E \times F, \mathcal{E} \otimes \mathcal{F}, \mu)$ with coordinate projections $X(x, y) = x$ and $Y(x, y) = y$.
- By existence of RCDs on Borel spaces, take an RCD of Y given $\sigma(X)$; it is a kernel $\vartheta_{Y,\sigma(X)}(\omega, \cdot)$.
- Factor the $\sigma(X)$-measurable map $\omega \mapsto \vartheta_{Y,\sigma(X)}(\omega, B)$ through X to get a measurable $\kappa(x, B)$ (use the factorization lemma).
- Then $\kappa(x, \cdot)$ is a probability measure for μ_E-a.e. x; redefine on a null set to get it for all x.
- Finally, the RCD identity on sets $A \times F$ yields $\mu(A \times B) = \int_A \kappa(x, B)\, \mu_E(dx)$, and uniqueness follows from agreement on a countable generator + monotone class.

Theorem 3.7 (Snell envelope recursion and existence of an optimal rule)

Proof idea.

- Backward induction on time: at N you must stop, so $U_N = Z_N$.
- For the recursion, fix $T \in \mathcal{T}_{n-1}$ and split on $\{T = n-1\}$ vs. $\{T \geq n\}$; on the second event use the tower property and the definition of U_n to bound $\mathbb{E}[Z_T \mid \mathcal{F}_{n-1}]$ by $\max\{Z_{n-1}, \mathbb{E}[U_n \mid \mathcal{F}_{n-1}]\}$.
- Taking the essential supremum over T gives $U_{n-1} \leq$ RHS; equality comes from considering the two candidate strategies: stop now or continue optimally from n.
- Define $T_n^* = \min\{k \geq n : U_k = Z_k\}$; measurability of $\{T_n^* = k\}$ follows from adaptedness of U, Z.
- Optimality is proved by induction using the recursion: on the continuation set $\{T_n^* \geq n+1\}$ you have $U_n = \mathbb{E}[U_{n+1} \mid \mathcal{F}_n]$, and eventually $U_n = \mathbb{E}[Z_{T_n^*} \mid \mathcal{F}_n]$.

Theorem 3.10 (Dynamic programming / Bellman recursion)

Proof idea.

- Use induction to show the Snell envelope has the Markov form $U_n = V_n(X_n)$.
- Assume $U_n = V_n(X_n)$; then by the Snell recursion $U_{n-1} = \max\{g_{n-1}(X_{n-1}), \mathbb{E}[V_n(X_n) \mid \mathcal{F}_{n-1}]\}$.
- By the Markov property, $\mathbb{E}[V_n(X_n) \mid \mathcal{F}_{n-1}] = \mathbb{E}[V_n(X_n) \mid X_{n-1}] = \int_E V_n(y) Q(X_{n-1}, dy)$, which defines V_{n-1}.
- The optimal stopping time is the first entry into the stopping region $\{x : g_n(x) = V_n(x)\}$ (same hitting-time idea as in Theorem 3.7).

Theorem 3.14 (Martingale optimality principle)

Proof idea.

- Part (i) is the minimality property of the Snell envelope: by definition it dominates Z and is the smallest supermartingale doing so.
- For any stopping time T, optional sampling for supermartingales gives $\mathbb{E}[Z_T] \leq \mathbb{E}[U_T] \leq \mathbb{E}[U_0]$, proving (ii).
- For $T^* = \min\{n : U_n = Z_n\}$, the Snell recursion implies that U is a martingale up to T^*, hence $\mathbb{E}[U_{T^*}] = \mathbb{E}[U_0]$ and $U_{T^*} = Z_{T^*}$; therefore (iii).

Theorem 3.15 (Doob decomposition)

Proof idea.

- Define the compensator increments $\Delta A_n := Y_{n-1} - \mathbb{E}[Y_n \mid \mathcal{F}_{n-1}] \geq 0$; this is exactly the supermartingale drift.

- Let $A_0 = 0$ and $A_n = \sum_{k=1}^{n} \Delta A_k$; then A is predictable and increasing.
- Set $M_n := Y_n + A_n$ and check the martingale property by direct conditioning: $\mathbb{E}[M_n \mid \mathcal{F}_{n-1}] = Y_{n-1} + A_{n-1} = M_{n-1}$.
- Thus $Y = M - A$; uniqueness follows from uniqueness of conditional expectations / increments.

Theorem 3.21 (Threshold rule and asymptotics)

Proof idea.

- Reduce the secretary problem to a 1D Markov optimal stopping problem using the relative rank process X_n and the Snell/Bellman recursion for success probabilities.
- Show the continuation value $c_n = \mathbb{E}[U_{n+1} \mid \mathcal{F}_n^X]$ becomes an explicit average because X_{n+1} is conditionally uniform on $\{1, \ldots, n+1\}$ given past relative ranks (symmetry of random permutations).
- The stopping rule compares 'stop now' (only beneficial when $X_n = 1$) versus 'continue' (value c_n), yielding a threshold index m where the inequality flips.
- Asymptotics come from approximating sums by integrals, giving $m \approx N/e$ and success probability $\approx 1/e$.

Theorem 4.3 (Conditional Gaussianity: informal statement)

Proof idea.

- Exploit linear-Gaussian dynamics: (X_{n+1}, Y_{n+1}) is an affine function of (X_n) plus independent Gaussian noise.
- By induction, conditional on past observations $\mathcal{Y}_n$, the state X_n remains Gaussian (Gaussian family closed under affine maps and conditioning).
- Compute how the mean and covariance propagate: prediction via the linear dynamics, update via Gaussian conditioning (a regression formula).

Theorem 4.5 (Normal correlation / Gaussian regression with pseudo-inverse)

Proof idea.

- Write the joint vector (X, Y) as Gaussian with block covariance; compute the conditional mean as the L^2-best linear predictor (orthogonal projection in a Gaussian Hilbert space).
- Use the normal equations to solve for the regression matrix; when the covariance is not invertible use the Moore–Penrose pseudo-inverse to get the minimal-norm solution.
- Show the conditional covariance is the Schur complement (or its pseudo-inverse analogue), and verify the result by checking characteristic functions / completing the square in the Gaussian density.

Theorem 2.2 (Conditional Gaussianity)

Proof idea.

- Set up the linear state-space model and show (X_n, Y_n) is jointly Gaussian for each fixed n by affine transformations of Gaussians.
- Use that conditioning a multivariate Gaussian remains Gaussian with explicitly updated mean/covariance (given by regression).
- Conclude that the filtering distribution $\mathcal{L}(X_n \mid \mathcal{Y}_n)$ is Gaussian and therefore fully described by (m_n, Σ_n).

Theorem 5.4 (Equivalent disturbance representation)

Proof idea.

- Unroll the controlled dynamics to express X_n as a measurable function of initial state, policy decisions, and an i.i.d. disturbance sequence.
- Construct the disturbance space as a product space and define the controlled process as a coordinate map; verify it has the same transition kernel as the original model.
- This gives an equivalent 'canonical' representation where randomness is separated (noise) from control (policy), which simplifies measurability and dynamic programming arguments.

Theorem 5.13 (History dependence does not improve the value)

Proof idea.

- Compare a history-dependent policy with its 'Markovized' version: at each time n pick an action distribution depending only on the current state, equal to the conditional law of the original action given the state.
- Show by induction that both policies generate the same state distribution at all times (use the tower property and the transition kernel).
- Therefore they have the same expected total reward, so optimizing over Markov policies is sufficient.

Theorem 6.4 (Reward Iteration)

Proof idea.

- Define reward iteration under a fixed policy by repeatedly applying the one-step operator $T_n^{f_n}$ backward from time N.
- Prove by induction on n that the iterates equal the conditional expected remaining reward starting from state x at time n.
- The key step is the tower property + Markov property to replace conditioning on full history by conditioning on the current state.

Theorem 6.6 (Verification Theorem)

Proof idea.

- Take a candidate function sequence (v_n) satisfying the Bellman inequalities $v_n \geq T_n v_{n+1}$ and terminal condition; show by backward induction that v_n dominates the value function V_n (monotonicity of T_n).
- If equality holds and maximizers exist, build a policy choosing stage-wise maximizers; then reward iteration shows this policy attains v_n.
- Conclude $v_n = V_n$ and the constructed policy is optimal.

Theorem 6.8 (Structure Theorem)

Proof idea.

- Backward induction using the structure assumption: start with $V_N = g_N \in M_N$.
- Assume $V_{n+1} \in M_{n+1}$; then closure (SAN-ii) gives measurability of $T_n V_{n+1}$ and keeps you in M_n.
- Upper bound: any policy value satisfies $V_n^\pi \leq T_n V_{n+1}$, hence $V_n \leq T_n V_{n+1}$ after taking the supremum.
- Lower bound: pick a maximizer f_n^* for V_{n+1} (SAN-iii) and patch with ε-optimal continuation to show $V_n \geq T_n V_{n+1}$.
- Collect the maximizers over all times and apply the verification theorem to get an optimal policy.

Theorem 7.15 (Daniell–Kolmogorov)

Proof idea.

- Define a pre-measure on cylinder sets using the consistent finite-dimensional distributions.
- Show this pre-measure is countably additive on the algebra of cylinder sets (consistency is crucial).
- Apply Carathéodory's extension theorem to extend uniquely to the product σ-algebra, obtaining a probability measure on the path space.
- Finally, the coordinate maps on the product space form the desired process with the given finite-dimensional laws.

Theorem 7.18 (Existence of stochastic processes in Lusin spaces)

Proof idea.

- Reduce the problem to the standard Borel/Polish case by representing the Lusin space as a Borel subset of a Polish space via a measurable isomorphism.
- Apply Daniell–Kolmogorov on the Polish model space to get a process there.
- Push the process forward through the isomorphism to obtain a process with the desired finite-dimensional distributions on the original Lusin space.

Theorem 8.1 (Existence (L´evy–Ciesielski / Haar–Schauder Construction)

Proof idea.

- Choose an orthonormal basis of $L^2[0,1]$ with good path properties (Haar functions) and define B_t as a random series with i.i.d. Gaussian coefficients (Lévy–Ciesielski).
- Use Kolmogorov's three-series / L^2 estimates to show the series converges uniformly a.s., giving continuous sample paths.
- Compute covariances to verify $\mathbb{E}[B_t] = 0$ and $\mathrm{Cov}(B_s, B_t) = \min\{s,t\}$, hence the finite-dimensional distributions are those of Brownian motion.
- Independence of increments follows from orthogonality of basis pieces corresponding to disjoint intervals.

Theorem 9.2 (Optional Stopping Theorem)

Proof idea.

- First prove the inequality for bounded stopping times by writing $\mathbb{E}[M_T] = \sum_n \mathbb{E}[M_n \mathbf{1}_{\{T=n\}}]$ and using the martingale property stepwise.
- For general integrable/bounded-below martingales, approximate T by $T \wedge n$ and pass to the limit using dominated/monotone convergence plus uniform integrability assumptions.
- The key idea is: martingales have zero conditional drift, so stopping cannot create expected gain unless integrability fails.

Theorem 9.3 (Skorokhod embedding)

Proof idea.

- Construct a stopping time τ as the first exit time of Brownian motion from a carefully chosen interval depending on the target law μ.
- Use the martingale B_t and (optional stopping) on bounded approximations to ensure $\mathbb{E}[B_\tau] = 0$ and to match second moments.
- Choose the interval/randomization so that the distribution of B_τ matches μ exactly (typically via quantile or potential-theoretic arguments).
- Uniform integrability / bounded stopping-time approximations justify passing limits and conclude $B_\tau \sim \mu$.

Theorem 9.12 (Strong Markov property)

Proof idea.

- Use the exponential martingale $M_t(\theta) = \exp(i\theta B_t - \frac{1}{2}\theta^2 t)$ and optional stopping at $T \wedge n$ to compute conditional characteristic functions of increments after T.
- Let $n \to \infty$ and use continuity of paths + dominated convergence to pass from $T \wedge n$ to T.
- The conditional characteristic function of $B_{T+t} - B_T$ given $\mathcal{F}_T$ equals $\exp(-\frac{1}{2}\theta^2 t)$, hence the increment is $N(0, t)$.
- Because this conditional characteristic function is deterministic, the increment is independent of $\mathcal{F}_T$; iterate to obtain independent stationary increments after T.

Theorem 9.14 (Reflection principle)

Proof idea.

- For $a > 0$, relate paths that hit level a before time t to paths whose endpoint exceeds a by reflecting the trajectory after the first hitting time τ_a.
- Show the reflection map preserves Wiener measure and swaps the events $\{\sup_{s \le t} B_s \ge a,\ B_t < b\}$ and $\{B_t > 2a - b\}$.
- This yields $\mathbb{P}(\sup_{s \le t} B_s \ge a) = 2\mathbb{P}(B_t \ge a)$ and more general joint identities for $(\sup B_s, B_t)$.

Theorem 10.18 (Relation E to M (E)

Proof idea.

- Show that integration against bounded continuous test functions determines a probability measure on a Polish space (separating class).
- Use regularity and a monotone class argument to pass from indicators of open/closed sets to bounded continuous functions, connecting set-based and function-based descriptions.
- Conclude that the embedding $\mu \mapsto \left(\int f\, d\mu\right)_{f \in C_b(E)}$ is injective and metrizable via the Prokhorov metric framework.

Theorem 10.19 (Portmanteau theorem and Prokhorov metric)

Proof idea.

- ($\Rightarrow$) If $\mu_n \Rightarrow \mu$, approximate indicators of open/closed sets by bounded continuous functions from below/above to get the liminf/limsup inequalities.
- ($\Leftarrow$) Conversely, use the inequalities for all closed sets (or a convergence-determining class) to show convergence of integrals for bounded continuous functions via approximation.
- Relate these conditions to the Prokhorov metric by showing $d_P(\mu_n, \mu) \to 0$ iff the portmanteau inequalities hold.

Theorem 11.4 (Prokhorov's theorem)

Proof idea.

- Use tightness to extract a subsequence that is Cauchy in the Prokhorov metric by a diagonal argument over a countable determining class (e.g. bounded Lipschitz functions).
- Show completeness/compactness: on a Polish space, tightness gives relative compactness of $\{\mu_n\}$ and any limit point is a probability measure.
- Conversely, if $\{\mu_n\}$ is relatively compact, show it must be tight; otherwise mass would escape to infinity and prevent weakly convergent subsequences.

Theorem 11.5 (Prokhorov theorem on metric spaces)

Proof idea.

- Adapt Prokhorov's theorem from Polish spaces to general metric spaces by working on the completion or by restricting to the support of tight measures.
- Show that tightness still prevents mass escape and yields sequential compactness in the weak topology (often for Radon measures).
- Conversely, weak relative compactness forces tightness by testing on complements of large compact sets and using the portmanteau theorem.

Theorem 11.9 (Tightness criterion based on Arzel'a–Ascoli)

Proof idea.

- Use Arzelà–Ascoli: relative compactness in $C([0,T];E)$ follows from (i) tightness at a fixed time (or at 0) and (ii) uniform control of the modulus of continuity.
- Translate (ii) into a probabilistic equicontinuity condition: for each ε choose δ so that increments over intervals of length δ are small with high probability, uniformly in n.
- Combine with a diagonal/compactness argument to get tightness of the laws on path space.

Theorem 11.16 (Kolmogorov–Chentsov)

Proof idea.

- Assume moment bounds on increments: $\mathbb{E}[d(X_t, X_s)^\alpha] \le C|t-s|^{1+\beta}$.
- Apply Markov's inequality on dyadic grids and a Borel–Cantelli argument to show that, a.s., increments decay like $|t-s|^\gamma$ for some $\gamma < \beta/\alpha$.
- Upgrade the dyadic control to all times by chaining (triangle inequality), yielding a modification with γ-Hölder continuous paths.

Theorem 11.17 (Kolmogorov criterion for tightness)

Proof idea.

- Use Kolmogorov–Chentsov to get Hölder-continuous modifications under uniform increment moment bounds, which implies tightness in $C([0,T];E)$.
- Alternatively, combine moment bounds with the Arzelà–Ascoli tightness criterion by controlling the modulus of continuity in probability.
- Conclude that the family of process laws is tight on path space; hence weakly convergent subsequences exist.

Theorem 12.12 (Birkhoff (individual ergodic theorem)

Proof idea.

- Define $\overline{f} = \limsup S_n$ and $\underline{f} = \liminf S_n$ and first show they are T-invariant using $S_n \circ T - S_n \to 0$.
- Apply Hopf's maximal ergodic lemma to $h = f - g$ where g is any invariant integrable function: it yields $\mathbb{E}[(f-g)\mathbf{1}_{\{\sup_k (S_k(f)-g)>0\}}] \ge 0$, giving the key comparison inequality.
- Use this inequality twice to show $\int \overline{f}\, d\mathbb{P} \le \int g\, d\mathbb{P}$ for invariant majorants g and $\int \underline{f}\, d\mathbb{P} \ge \int g\, d\mathbb{P}$ for invariant minorants; choosing $g = \underline{f}$ and $g = \overline{f}$ forces $\overline{f} = \underline{f}$ a.s.
- Let $f^* = \lim S_n$; it is invariant, hence $\mathcal{I}$-measurable. Identify it by testing on invariant sets $A \in \mathcal{I}$: Hopf's inequality applied to $f\mathbf{1}_A$ gives $\int_A f^* = \int_A f$, which characterizes $\mathbb{E}[f \mid \mathcal{I}]$.
- L^1 convergence follows by dominated convergence for bounded f and truncation plus L^1-contraction for general $f \in L^1$.